

# Syslog

## Project Overview

This project explores the configuration and security of the Syslog protocol for centralized log management in network environments. Key objectives include:

- Understand Syslog Functionality  
Learn how Syslog transmits log messages from devices to a central server.
- Analyze Syslog Message Format  
Review components: priority, timestamp, hostname, and message text.
- Configure Syslog on Cisco Devices  
Use commands like logging, trap, and service timestamps for setup.
- Identify Affected Network Components  
Includes:
  - Switches, virtual switches, access points
  - Wireless controllers, DMZ, edge devices
  - Untrusted ports
- Address Security Concerns  
Cover risks like plain-text transmission, spoofing, message tampering, and DoS.
- Apply Best Practices  
Use TCP/TLS, enable authentication, restrict access, and monitor logs regularly.

## Description

Syslog is a standardized protocol used for transmitting log messages in computer systems, particularly from network devices to a central log server. It is widely used for event logging, error messages, diagnostics, and auditing purposes. Syslog protocol is simple, flexible, and supported by many devices and platforms.

## Recommendations

- Set a central syslog server
- Use timestamps
- Set Logging levels wisely
- Use hostnames for Readability
- Enable buffer logging
- Monitor logs in real-time (Optional)
- Protect syslog communication
- Log filtering
- Use SNMP or Alerts for Critical Logs
- Regular Log Rotation and Backup

## References

<https://www.geeksforgeeks.org/computer-networks/what-is-syslog-server-and-its-working/>  
<https://wirexsystems.com/resource/protocols/syslog>  
<https://www.crowdstrike.com/en-us/guides/syslog-logging/>  
<https://unofficialaciguide.com/2018/08/11/configuring-syslog-for-aci/>

## Details

All network devices such as routers, servers, firewalls, etc. create or prompt logs about statuses and the events that occur. For a small system tracking these logs is not a problem, the problem arises when we are dealing with big systems where tracking all these logs and information becomes challenging. To overcome this problem we use Syslog with a logging server known as Syslog server (such as Kiwi Syslog server, Graylog, Solarwind Syslog server, etc.).

A Syslog server allows us to send the log information of all our network devices to one centralized place. The log messages are sent on UDP port 514 to the Syslog server. From here we can search, manage and archive all of the log information. Syslog typically uses the User Datagram Protocol (UDP) for transport, with port 514 being the default port. However, it can also use the Transmission Control Protocol (TCP) or other transport protocols for more reliable delivery at the expense of additional overhead.

### Syslog has three main components:

1. Syslog Client (Sender): A device or application that generates log messages. Syslog clients can include servers, routers, switches, firewalls, or other network devices and applications that produce log information.
2. Syslog Server (Receiver): A centralized server responsible for receiving and storing log messages from multiple clients. The server may offer filtering, analysis, and reporting functionalities to help administrators manage and monitor their networks.
3. Syslog Message: The log message itself, following a specific format consisting of a priority value, timestamp, hostname, and the actual message text.

### Syslog messages have three parts:

1. Priority: An integer value representing the severity and facility of the message. Severity levels range from 0 (emergency) to 7 (debug). Facilities identify the source or type of the message, such as kernel messages, user-level messages, or mail system messages.
2. Header: Contains the timestamp and hostname (or IP address) of the device generating the log message.
3. Message: The actual log message text, detailing the event or issue being logged.

Syslog message format:

```
seq:timestamp: %facility-severity-MNEMONIC:description
```

1. **seq** - A sequence number indicating the sequence/order of a message.
2. **timestamp** - A timestamp indicates at what time the message was generated.
3. **facility** - A value that indicates which process on the device generated this message.
4. **severity** - A number that indicates the severity of the logged event. There are 8 severity levels.
5. **MNEMONIC** - A shortcode for the message, indicating what happened.
6. **description** - Detailed information about the event being reported.

### Syslog Severity levels:

1. **Level 0:** Emergency, events that render the system unusable.
2. **Level 1:** Alert, is for events for which action must be taken immediately. So, these are also very urgent/serious events.
3. **Level 2:** Is called critical, and the description is simply 'critical conditions'. Same
4. **Level 3:** Error.
5. **Level 4:** Warning.
6. **Level 5:** Notice/notification, is used for messages representing a 'normal but significant condition'.
7. **Level 6:** Is 'Informational', and then finally
8. **Level 7:** Is Debugging. These are the least severe messages.

### Syslog Security Concerns

1. Unencrypted Transmission
  - Standard Syslog sends logs in plain text, making them vulnerable to eavesdropping.
2. No Authentication
  - Lacks built-in client-server authentication, allowing spoofed or fake log entries.
3. Message Tampering
  - No integrity checks mean logs can be altered during transit.
4. Unreliable Transport
  - Uses UDP by default, which can lose or reorder messages.
5. DoS Attacks
  - Syslog servers can be overwhelmed by log floods, disrupting operations.

### Syslog Configuration (Cisco Devices)

#### Basic Setup

- *logging <Syslog-Server-IP>*  
*Send logs to central server*
- *logging trap warnings*  
*Set minimum severity level (0-7)*
- *service timestamps log datetime msec*  
*Add date/time to logs*
- *logging buffered 4096 (optional)*  
*Store logs locally in device buffer*
- *no logging console*  
*Prevent console log spam*
- *logging monitor warnings*  
*Show logs during remote sessions*